

Relations between the sides and the angles of a triangle

The longer side faces the larger angle — and the triangle inequality follows from the same idea.

LESSON 92–93

2 × 40 MIN

MATEMATIKA 7, §43

S8 5

LESSON CLOCK

15'

20'

25'

20'

5'

The larger angle faces the longer side	15 min
Reading the order both ways	20 min
The triangle inequality	25 min
The shortest path to a line	20 min
Homework	5 min

LEARNING OBJECTIVES

- State that the larger angle faces the longer side, and its converse.
- Order the sides of a triangle from its angles, and the angles from its sides.
- Use the triangle inequality to decide whether three lengths make a triangle.
- Show that the perpendicular is the shortest segment from a point to a line.

TEXTBOOK

National	pp. 269–276
Cambridge	Stage 8, pp. 44–56
Workbook	Exercise 5.4

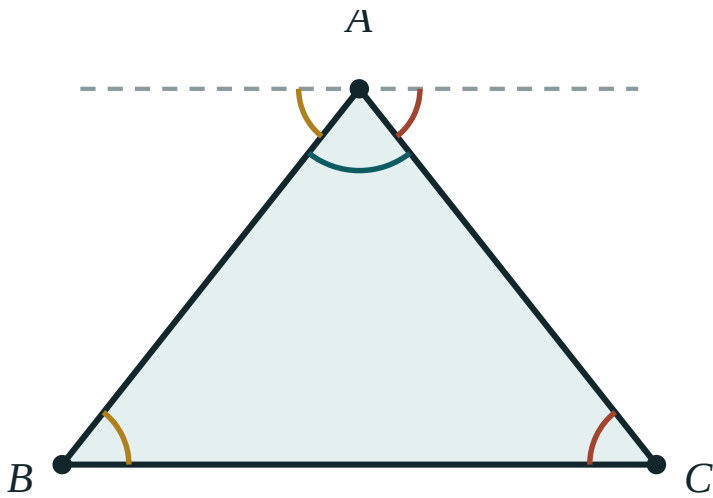
01

Explanation

The larger angle faces the longer side

In any triangle, the longer of two sides is opposite the larger of two angles — and the statement runs backwards as well.

$$AB > AC \iff \angle C > \angle B$$



the three angles fill 180°

Each side faces the angle of matching rank

TRIANGLE	LARGEST ANGLE	LONGEST SIDE
$\angle A = 90^\circ, \angle B = 60^\circ, \angle C = 30^\circ$	$\angle A$	BC
$\angle A = 40^\circ, \angle B = 40^\circ, \angle C = 100^\circ$	$\angle C$	AB
equilateral	all equal	all equal

A SIDE AND THE ANGLE AT ITS OWN END ARE NOT PARTNERS

The side BC is matched with $\angle A$, the angle it faces, not with $\angle B$ or $\angle C$. Reading the label off the wrong vertex reverses every answer.

Reading the order both ways

Ranking the angles ranks the sides in exactly the same order, and the other way round.

GIVEN	ORDER OF ANGLES	ORDER OF SIDES
$\angle A = 80^\circ, \angle B = 60^\circ, \angle C = 40^\circ$	$\angle A > \angle B > \angle C$	$BC > AC > AB$
$a = 5, b = 7, c = 9$	$\angle C > \angle B > \angle A$	$c > b > a$
isosceles with $AB = AC$	$\angle B = \angle C$	$AC = AB$

THE ISOSCELES THEOREM IS THIS THEOREM

Equal sides $\iff$ equal base angles is the borderline case of the same statement: when neither side is longer, neither angle is larger.

The triangle inequality

Each side of a triangle is shorter than the sum of the other two — the straight path from one vertex to another beats any detour through a third.

$$a < b + c, b < a + c, c < a + b$$

LENGTHS	CHECK THE LONGEST	A TRIANGLE?
3, 4, 5	$3 + 4 > 5$	yes
5, 8, 14	$5 + 8 < 14$	no
6, 9, 13	$6 + 9 > 13$	yes
2, 3, 5	$2 + 3 = 5$	no — the three points are in a line

ONLY THE LONGEST SIDE NEEDS CHECKING

If the longest side is beaten by the sum of the other two, the other two inequalities hold automatically. One line of working settles the question.

The shortest path to a line

Drop a perpendicular MP from a point M to a line, and take any other segment MQ to that line.

STATEMENT	REASON
$\angle MPQ = 90^\circ$	construction
$\angle MPQ$	the largest angle of $\triangle MPQ$
$MQ > MP$	the longer side faces the larger angle

So the perpendicular is shorter than every slant segment — which is exactly why distance to a line was defined along the perpendicular two lessons ago.

A DEFINITION JUSTIFIED AFTER THE EVENT

Calling the perpendicular “the distance” was a choice; this theorem shows it was the only sensible one, since it is the minimum.

02 Worked examples

EXAMPLE 1 In $\triangle ABC$, $\angle A = 80^\circ$, $\angle B = 60^\circ$. Order the sides.

① $\angle C = 180^\circ - 140^\circ = 40^\circ$

② $\angle A > \angle B > \angle C$

③ Each side faces the angle of matching rank.

④ $BC > AC > AB$

ANSWER

$BC > AC > AB$

EXAMPLE 2 Can 5, 8, 14 be the sides of a triangle? And 6, 9, 13?

① Longest of the first is 14.

② $5 + 8 = 13 < 14$ — no.

③ Longest of the second is 13.

④ $6 + 9 = 15 > 13$ — yes.

ANSWER

No; yes

EXAMPLE 3 Two sides of a triangle are 7 and 10. Between what values must the third lie?

- ① It must beat the difference: $x > 10 - 7 = 3$.
- ② It must be beaten by the sum: $x < 10 + 7 = 17$.
- ③ $3 < x < 17$

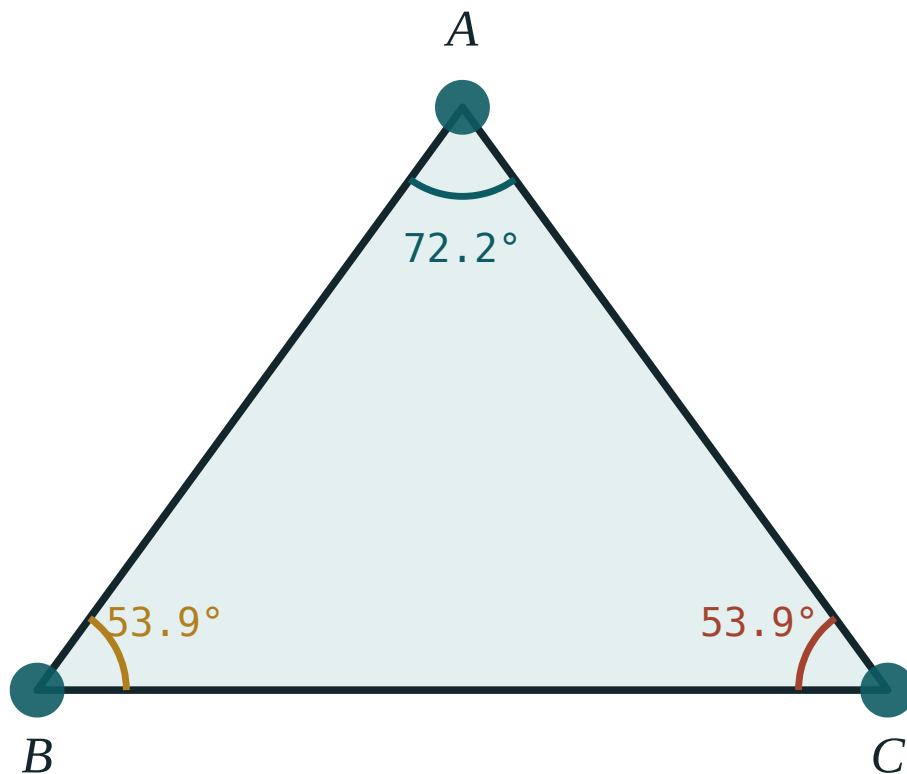
ANSWER

$$3 < x < 17$$

03 Interactive model

Hand out strips of 5, 8 and 14 cm and ask the class to build the triangle; the failure is visible and the inequality is never forgotten.

Watch the longest side and the largest angle move together



$\angle A$ 72.2°

$\angle B$ 53.9°

$\angle C$ 53.9°

sum 180°

longest side **BC**

largest angle $\angle A$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Opposite side	Qarshi tomon	Противолежащая сторона
The larger	Kattaroq	Больший
The smaller	Kichikroq	Меньший
To order	Tartiblash	Упорядочить
Triangle inequality	Uchburchak tengsizligi	Неравенство треугольника
Perpendicular	Perpendikulyar	Перпендикуляр
Slant segment	Og'ma	Наклонная
Shortest	Eng qisqa	Кратчайший

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The longest side faces:

the smallest angle

the largest angle

the right angle only

any angle

In a right-angled triangle the longest side is:

a leg

the hypotenuse

either

undetermined

$\angle A = 80^\circ$, $\angle B = 60^\circ$, $\angle C = 40^\circ$ gives:

$AB > BC > AC$

$BC > AC > AB$

$AC > AB > BC$

all equal

Sides 5, 8, 14 make:

a triangle

no triangle

a right triangle

an isosceles triangle

To test three lengths you need to check:

all three sums

the longest side only

the shortest side

the perimeter

The shortest segment from a point to a line is:

any slant

the perpendicular

the bisector

the median

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The longest side faces the

ANSWER largest angle

2. The shortest side faces the

ANSWER smallest angle

3. 3, 4, 5: a triangle?

ANSWER Yes

4. 5, 8, 14: a triangle?

ANSWER No

5. 2, 3, 5: a triangle?

ANSWER No

6. In a right triangle the longest side is the

ANSWER hypotenuse

7. The shortest segment to a line is the

ANSWER perpendicular

Medium · the standard question

7 PROBLEMS

1. $\angle A = 80^\circ$, $\angle B = 60^\circ$: order the sides

ANSWER $BC > AC > AB$

2. $a = 5$, $b = 7$, $c = 9$: order the angles

ANSWER $\angle C > \angle B > \angle A$

3. Sides 7 and 10: the range of the third

ANSWER $3 < x < 17$

4. Sides 4 and 9: the range of the third

ANSWER $5 < x < 13$

5. An obtuse triangle: which side is longest?

ANSWER The one facing the obtuse angle

6. Isosceles with $AB = AC$: which angles are equal?

ANSWER $\angle B = \angle C$

7. 6, 9, 13: a triangle?

ANSWER Yes

Hard · stretch and reason

7 PROBLEMS

1. The perimeter of a triangle with sides 7, 10, x is a whole number: how many values of x are whole?

ANSWER 13 — from 4 to 16

2. Prove that the perpendicular is shorter than any slant segment

ANSWER It faces the largest angle of the right triangle

3. In $\triangle ABC$, $\angle A = 90^\circ$: prove BC is the longest side

ANSWER It faces the largest angle

4. Two sides are 8 and 15 and the third is a whole number: the smallest possible

ANSWER 8

5. Can a triangle have sides 1, 2, 3?

ANSWER No — $1 + 2 = 3$

6. If $\angle A > \angle B$ and $\angle B > \angle C$, order the sides

ANSWER $a > b > c$

7. Why is checking the longest side enough?

ANSWER The other two inequalities then hold automatically

07 Homework

Homework — 5 tasks

For a triangle test, find the longest side first and check only that one.

- In $\triangle ABC$, $\angle A = 95^\circ$ and $\angle B = 45^\circ$. Order the three sides.
- Decide whether 4, 6, 11 and 4, 6, 9 make triangles.
- Two sides are 9 and 12. Find the range of the third.
- Explain in two lines why the perpendicular is the shortest path to a line.

5. A triangle has $a = 6$, $b = 6$, $c = 10$. Order its angles.